Charles Fucile

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Education

Georgia Institute of Technology Aug 2024 – May 2027
Bachelor of Science in Aerospace Engineering – *John H. Martinson Honors Program* GPA: 3.90
Relevant Coursework: Aerodynamics, Structural Analysis, Thermodynamics, Computational Fluid Dynamics, Jet & Rocket Propulsion, Orbital Mechanics, Dynamics, Experimental Methods, Circuits & Electronics

Technical Experience

Computational Fluid Dynamics Research Jan 2026 – Present
Georgia Institute of Technology

- Conducted CFD simulations in Ansys Fluent across four flow configurations: symmetric airfoil at varying angles of attack, internal flow through a converging-diverging nozzle, boundary layer development over a flat plate, and arterial flow with and without plaque occlusion.
- Generated structured and unstructured meshes, performed mesh refinement studies, and verified solution convergence to ensure numerical validity across all cases.
- Quantified pressure distributions, velocity fields, and force coefficients; compiled comparative tabulated analyses and technical presentations for each configuration.

Yellow Jacket Space Program – Launch Fluids Engineer Aug 2024 – Dec 2025
Georgia Tech Liquid Rocketry Team

- Assembled, maintained, and operated high-pressure propellant and pressurization fluid systems supporting rocket ground test operations.
- Diagnosed and eliminated fuel line leaks by disassembling and resealing the fuel pressurization system, restoring full test readiness.

Projects

Multi-Body Orbital Dynamics Simulation Dynamics, Georgia Tech

- Developed MATLAB simulations of the three-body gravitational problem to map Lagrange point equilibria and model binary star system formation dynamics within a team environment.
- Numerically integrated equations of motion across a range of mass ratios and initial conditions to characterize system stability and orbital behavior.

Zero-G Cubic Robot Simulation & Design Dynamics, Georgia Tech

- Independently built a MATLAB rigid-body simulation of a cubic robot operating in a microgravity space station environment, modeling attitude control and motion constraints.
- Applied simulation results to directly inform preliminary design decisions for the robot's mechanical configuration and control approach within a team design effort.

Leadership

AeroMaker Space – Student Mentor Jan 2026 – Present

- Mentored 200+ students across a high-traffic makerspace with 20+ machines, providing hands-on support for additive manufacturing (FDM, SLA, SLS), laser cutting, and CNC workflows.
- Performed equipment maintenance and repair including full restoration of a band saw and ongoing repair of a table saw; managed 3D printer slicing and job queuing for rapid prototyping operations.

Skills

Engineering: Aerodynamics, CFD Analysis, Systems Integration, Test Operations, Mechanical Assembly, Prototyping
Software: Ansys Fluent, MATLAB, SolidWorks, Fusion 360, Onshape
Manufacturing: Additive Manufacturing (FDM, SLA, SLS), Laser Cutting, CNC Milling & Lathing, Equipment Maintenance & Repair, Soldering